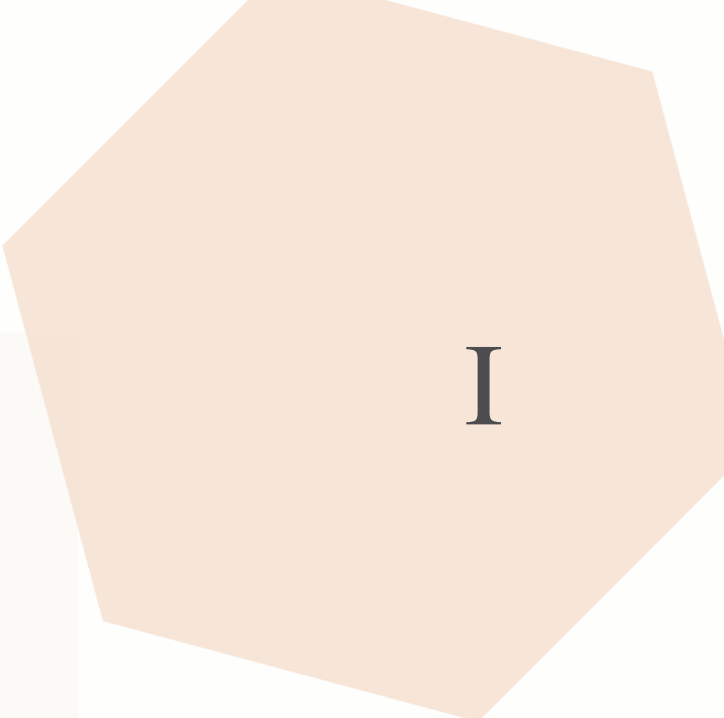


Measure Theory and Lebesgue Integration

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Introduction

This article introduces the foundations of measure theory and the Lebesgue integral, which together form the backbone of modern real analysis. We begin with σ -algebras and measures, construct the Lebesgue measure on $\mathbb{R}^n$, and develop the integral with respect to a general measure.

1 Sigma-Algebras and Measurable Spaces

Let X be a non-empty set. A collection $\mathcal{F} \subseteq \mathcal{P}(X)$ is a σ -algebra on X if:

- (i) $X \in \mathcal{F}$;
- (ii) if $A \in \mathcal{F}$, then $A^c \in \mathcal{F}$;
- (iii) if $(A_n)_{n=1}^{\infty}$ is a sequence in $\mathcal{F}$, then $\bigcup_{n=1}^{\infty} A_n \in \mathcal{F}$.

The pair $(X, \mathcal{F})$ is called a *measurable space*.

The Borel σ -algebra $\mathcal{B}(\mathbb{R})$ is the σ -algebra generated by the open sets of $\mathbb{R}$. It contains all open sets, closed sets, countable intersections of open sets (G_δ sets), and countable unions of closed sets (F_σ sets).

2 Measures

Let $(X, \mathcal{F})$ be a measurable space. A function $\mu: \mathcal{F} \rightarrow [0, \infty]$ is a *measure* if:

- (i) $\mu(\emptyset) = 0$;
- (ii) (countable additivity) for any sequence (A_n) of pairwise disjoint sets in $\mathcal{F}$,

$$\mu\left(\bigcup_{n=1}^{\infty} A_n\right) = \sum_{n=1}^{\infty} \mu(A_n).$$

Let μ_0 be a pre-measure on an algebra $\mathcal{A} \subseteq \mathcal{P}(X)$. Then there exists a measure μ on $\sigma(\mathcal{A})$ that extends μ_0 . If μ_0 is σ -finite, the extension is unique.

Proof. Define the outer measure $\mu^*(E) = \inf \sum_n \mu_0(A_n)$, where the infimum is taken over all countable covers $\{A_n\} \subseteq \mathcal{A}$ of E . One then shows that the collection of μ^* -measurable sets forms a σ -algebra containing $\mathcal{A}$, and μ^* restricted to this σ -algebra is a measure extending μ_0 . $\square$

3 The Lebesgue Integral

Let $(X, \mathcal{F}, \mu)$ be a measure space and $\varphi = \sum_{i=1}^n a_i \mathbf{1}_{A_i}$ a non-negative simple function. The Lebesgue integral of φ is

$$\int_X \varphi d\mu = \sum_{i=1}^n a_i \mu(A_i).$$

Let (f_n) be a sequence of non-negative measurable functions with $f_n \uparrow f$ pointwise. Then

$$\lim_{n \rightarrow \infty} \int_X f_n d\mu = \int_X f d\mu.$$

The Monotone Convergence Theorem allows us to exchange limits and integrals for increasing sequences. This is the engine that powers most constructions of the Lebesgue integral: approximate from below by simple functions and pass to the limit.

Let (f_n) be a sequence of measurable functions converging pointwise to f . If there exists an integrable function g such that $|f_n| \leq g$ for all n , then

$$\lim_{n \rightarrow \infty} \int_X f_n d\mu = \int_X f d\mu.$$

The domination condition $|f_n| \leq g$ with g integrable cannot be dropped. The sequence $f_n = n \cdot \mathbf{1}_{(0,1/n)}$ converges pointwise to 0 on $(0, 1)$, yet $\int f_n = 1$ for all n .